

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: AI & Machine Learning

Challenge ID: PS-07

Title: CYBERGUARD – AI-Powered Cyber Threat & Phishing Detection

1. Executive Summary & Problem Overview

Modern cyber threats have evolved beyond crude spam emails into sophisticated, AI-augmented social engineering campaigns. Organizations and individual internet users face relentless vectors of attack, including zero-day spear-phishing, homograph lookalike domain spoofs, malicious credential-harvesting links, compromised enterprise credentials, and suspicious lateral network movements.

Traditional signature-based antivirus solutions and static rule filters are fundamentally reactive; they struggle against obfuscated URLs, dynamically generated domains, multi-stage redirect chains, and weaponized natural language lures. Security Operations Center (SOC) teams are inundated by alert fatigue and lack explainable, context-rich risk scoring to make rapid containment decisions.

The objective of **PS-07 (CYBERGUARD)** is to design and develop an intelligent, explainable cybersecurity analytics platform. The system must ingest and analyze emails, suspicious URLs, text messages, authentication access logs, and network telemetry to detect phishing, credential theft, executive impersonation, and anomalous network behaviors, producing actionable, explainable threat scores and automated incident response recommendations.

2. Core Personas & Access Roles

1. **SOC Analyst / Incident Responder:** Analyzes flagged threat alerts, reviews explainable forensic breakdowns, investigates malicious payloads, and executes automated isolation workflows.
2. **Enterprise IT & Security Administrator:** Manages security telemetry ingestion policies, configures organizational risk thresholds, reviews identity access anomalies, and audits compliance posture.
3. **End-User / Corporate Employee:** Submits suspicious emails or links via an interactive scanner or browser plugin, receiving real-time safety scores and phishing awareness tips.
4. **CISO / Executive Leadership:** Monitors high-level enterprise threat metrics, trending attack vectors, mean-time-to-detect (MTTD), and overall cyber resilience health.

3. Key Functional Modules & Technical Deliverables

Module 1: Intelligent Phishing & Email Lure Analysis

- **NLP-Driven Intent & Sentiment Extraction:** Scans email headers, body text, and attachments to identify deceptive urgency, financial extortion cues, impersonation of authority figures, and credential-solicitation language.
- **Header & Domain Authentication Forensics:** Inspects SPF, DKIM, and DMARC alignments, analyzing lookalike/typosquatted sender domains (e.g., Unicode homoglyphs, deceptive TLDs).
- **Attachment Inspection & Payload Heuristics:** Safely parses email attachment metadata and macro signatures to flag disguised executables and malicious scripts.

Module 2: Real-Time Malicious URL & Web Threat Inspector

- **Lexical & Structural URL Forensics:** Analyzes link patterns, entropy scores, suspicious subdomains, shortening services, and abnormal path structures.
- **Redirection Chain & Sandbox Simulation:** Follows HTTP redirect hops to the final destination, scanning SSL certificate validity, domain age (WHOIS), and registrar reputation.
- **Visual & Content Clone Detection:** Compares the destination web page against known brand logos, login portals, and layout templates to catch zero-day credential-phishing replicas.

Module 3: Authentication Log & Behavioral Anomaly Engine

- **Access Log Analytics:** Parses authentication records (e.g., auth.log, Active Directory, AWS CloudTrail, or simulated Syslog streams) to identify suspicious activities.
- **Impossible Travel & Credential Stuffing Detection:** Flags simultaneous logins across geographically distant IP blocks, rapid brute-force spikes, and abnormal off-hours access.
- **Privilege Escalation & Lateral Movement Tracking:** Uses unsupervised machine learning to detect user account deviations from baseline behavioral profiles.

Module 4: Explainable Risk Scoring & Automated Response (SOAR)

- **Explainable Threat Attribution (XAI):** Provides transparent, human-understandable justification for every flagged threat (e.g., *"Flagged 92/100: Mismatched DMARC, homoglyph domain, and high-urgency credential request detected"*), avoiding black-box ambiguity.
- **MITRE ATT&CK Mapping:** Classifies detected threats against standardized cyber kill-chain techniques (e.g., T1566: Phishing, T1078: Valid Accounts).
- **Automated Containment Playbooks:** Generates recommended immediate remediation steps (e.g., auto-quarantining malicious emails, generating firewall IP blocklist rules, triggering password resets).

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web / SOC Console, React.js (Vite) / Next.js / Tailwind CSS / Lucide Icons / Recharts

Browser Extension, Chrome / Firefox Manifest V3 Extension for real-time link/email scanning

Backend & Ingestion, Python (FastAPI / Flask) or Node.js (Express) with REST APIs

Database & Caching, PostgreSQL / Supabase + Redis for fast threat intelligence caching

AI / Machine Learning, "Python (Scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, RoBERTa / DeBERTa for phishing NLP, SHAP / LIME for explainability)"

Security & Threat Feeds, "OpenPhish / PhishTank public datasets, VirusTotal API, WHOIS / DNS lookup libraries, MaxMind GeoIP"

Telemetry Parsing, Elasticsearch / Logstash or Python log parsers for authentication events

5. Evaluation & Judging Criteria (100 Points Total)

1. **Detection Accuracy & Multi-Vector Coverage (25 Points):** Robustness across multiple attack vectors (emails, URLs, authentication logs, and network behavior).
2. **Explainability & Threat Justification (25 Points):** Clarity and transparency of the AI reasoning engine; providing actionable forensic rationales rather than raw unverified scores.
3. **Automated Response & Remediation Logic (20 Points):** Practical utility of generated playbooks, blocklist generation, and incident containment workflows.
4. **SOC Interface & User Experience (15 Points):** Clean analyst console, intuitive threat severity badges, smooth link submission, and real-time dashboard responsiveness.
5. **Code Quality, Sprint Execution & Standards (15 Points):** Well-structured GitHub repository, sample test datasets (benign vs. phishing samples), .env.example, and a working live prototype.

6. Submission Guidelines

1. **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup guide, sample benign and phishing datasets, and .env.example.
2. **Live Deployment:** Hosted interactive web portal or SOC dashboard (Vercel / Render / AWS).
3. **Demo Video / Deck:** A crisp 3-minute video showing:
 4. Submission and scanning of a phishing email or malicious URL.
 5. Live display of explainable risk scores and attribution markers.
 6. Ingestion of authentication logs triggering an impossible travel / brute-force anomaly.
 7. Generation of an automated incident containment recommendation.